Laboratory 03

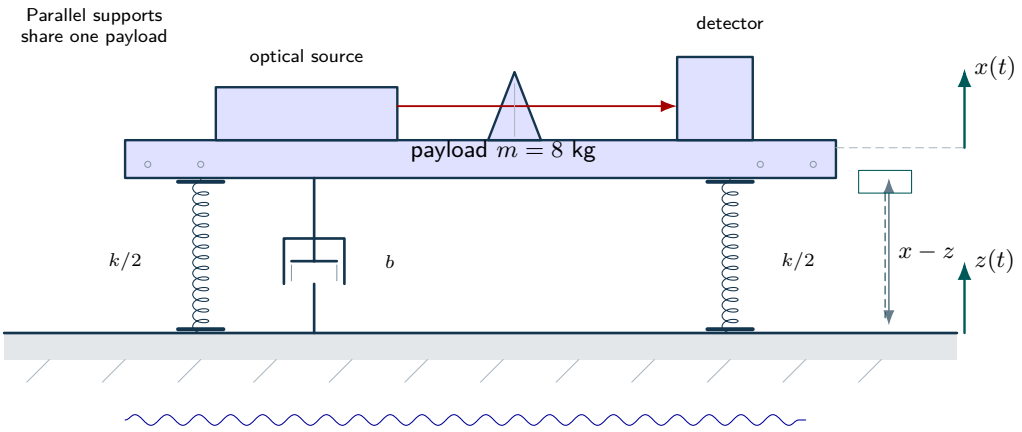
A precision platform under floor motion

Exercise sheet – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Nominal payload: $m = 8$ kg, $k = 3200$ N/m and $\zeta = 0.08$ unless varied. Absolute payload motion is x; floor motion is z. Spring and damper depend on relative motion about static equilibrium.

1. An optical platform on a moving foundation

A precision payload is supported by a spring and damper attached to a moving floor. Positions are measured about the loaded static equilibrium, so gravity does not appear in the incremental motion equation.



Measured coordinates

x, z: absolute displacements [m].
$x - z$: spring travel [m].
$\dot{x} - \dot{z}$: damping velocity.

Force balance

$m\ddot{x} + b(\dot{x} - \dot{z})$
$\quad +k(x - z) = 0.$
Two $k/2$ supports give total k.

Operating boundary

Coordinates are deviations about static sag. The blue floor trace indicates motion; it is not a measured waveform.

An optical payload rests on parallel spring and damping elements. The two springs each have stiffness $k/2$, giving total k. Both the floor and payload move; spring extension and damping velocity are relative quantities. Weight is balanced at the static equilibrium about which the model is written.

(a) An 8 kg platform is attached to a moving floor by a spring $k = 3200$ N/m and damper b. Let x and z be absolute platform and floor displacements. Derive X/Z and the relative-displacement transfer $(X - Z)/Z$.

(b) Find ω_n and the damping coefficients for ratios 0.08 and 0.4. Calculate the closed-form poles for both cases.

(c) Find the zero-frequency gains from floor displacement to absolute payload motion and to relative suspension travel. Explain the physical difference and why using the force-input transfer for this experiment would be wrong.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

2. Damping a resonance while preserving isolation

The design must cross a resonant region during operation but should isolate higher-frequency floor motion. Compare the competing effects of damping rather than ranking one damper as universally better.

(a) Using $r = \omega/\omega_n$, derive the condition for absolute displacement transmissibility below one. Explain why the nonzero threshold is independent of positive damping in this ideal model.

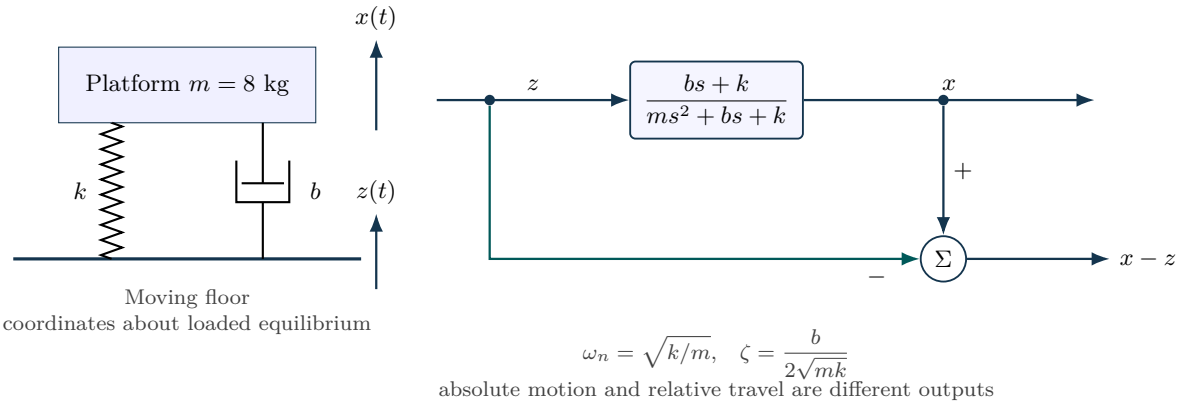
(b) Calculate transmissibility at $r = 1$ for damping ratios 0.08 and 0.4. Which design better suppresses floor motion at that frequency? Do these values necessarily equal the exact peak magnitudes?

(c) Analyze the limit $\zeta \to 0^+$ at $r = 1$ and at a fixed $r > \sqrt{2}$. Use the result to explain why an isolation threshold alone is an incomplete design specification.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

3. Absolute vibration and suspension-travel limits

The same floor motion can create different specifications for the optical payload and its supporting suspension. Evaluate both outputs before selecting a mount.



Base excitation drives absolute motion; subtracting the floor coordinate gives relative travel. The spring and damper act in parallel.

(a) Evaluate transmissibility at $r = 3$ for both damping ratios. For nonzero damping, derive the high-frequency asymptote and explain the design conflict with Problem 2(b).

(b) At $r = 2$, floor displacement amplitude is 0.5 mm. With $\zeta = 0.08$, find the amplitude of relative travel $x - z$ and decide whether it stays within a 0.8 mm amplitude limit.

(c) For a fixed floor-displacement amplitude of 0.5 mm, find the limiting absolute and relative amplitudes as frequency tends to infinity with nonzero damping. Relate these limits to the finite-frequency travel check.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

4. A heavier payload under a tight static-sag requirement

The payload may change, but static support and vibration isolation must still be achieved by the same spring. Treat these as simultaneous constraints.

(a) Keep $k = 3200$ and $b = 25.6$ but increase total mass to 12 kg. Find the new natural frequency, damping ratio and isolation threshold. Explain why keeping the damper fixed does not keep the damping ratio fixed.

(b) For the original 8 kg platform, calculate static spring deflection under gravity. If deflection may not exceed 3 mm, determine the minimum stiffness and the corresponding natural frequency, using $g = 9.81$ m/s^2.

(c) Require gravity sag at most 3 mm and isolation of floor motion at 40 rad/s. Derive mass-independent bounds on ω_n and decide whether any passive spring in this model satisfies both.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

5. What a decay test says about energy and excitation

A synthetic ring-down is used to infer damping. Distinguish that unforced test from operation on a moving floor, which can continually inject energy.

(a) In a noise-free synthetic free decay, consecutive positive peaks are 1.5 mm and 0.9 mm, separated by 0.32 s. Infer damping ratio, natural frequency and b if mass is 8 kg. State the assumption behind this inference.

(b) Set $z = 0$. Show that mechanical energy decreases for $b > 0$ and discuss asymptotic stability. Explain why the same autonomous energy argument cannot prove decreasing platform energy for arbitrary moving-floor excitation.

(c) For moving floor coordinate z, use physical stored energy $E = m\dot{x}^2/2 + k(x - z)^2/2$. Derive its power balance and identify the term that prevents an unconditional monotonic-decay claim.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.